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- **Term:** 3rd Year

Required Python Packages

Run the following applications once if they are not yet installed in your Python environment:

```
In [1]: !pip install numpy
```

Requirement already satisfied: numpy in c:\users\naval\anaconda3\lib\site-packages (2.3.5)

```
In [2]: !pip install scipy
```

Requirement already satisfied: scipy in c:\users\naval\anaconda3\lib\site-packages (1.16.3)
Requirement already satisfied: numpy<2.6,>=1.25.2 in c:\users\naval\anaconda3\lib\site-packages (from scipy) (2.3.5)

```
In [3]: !pip install matplotlib
```

Requirement already satisfied: matplotlib in c:\users\naval\anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (2.3.5)
Requirement already satisfied: packaging>=20.0 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\naval\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\naval\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

```
In [4]: !pip install panda
```

Requirement already satisfied: panda in c:\users\naval\anaconda3\lib\site-packages (0.3.1)
 Requirement already satisfied: setuptools in c:\users\naval\anaconda3\lib\site-packages (from panda) (80.9.0)
 Requirement already satisfied: requests in c:\users\naval\anaconda3\lib\site-packages (from panda) (2.32.5)
 Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\naval\anaconda3\lib\site-packages (from requests->panda) (3.4.4)
 Requirement already satisfied: idna<4,>=2.5 in c:\users\naval\anaconda3\lib\site-packages (from requests->panda) (3.11)
 Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\naval\anaconda3\lib\site-packages (from requests->panda) (2.6.3)
 Requirement already satisfied: certifi>=2017.4.17 in c:\users\naval\anaconda3\lib\site-packages (from requests->panda) (2026.1.4)

In [5]: !pip install gekko

Requirement already satisfied: gekko in c:\users\naval\anaconda3\lib\site-packages (1.3.2)
 Requirement already satisfied: numpy>=1.8 in c:\users\naval\anaconda3\lib\site-packages (from gekko) (2.3.5)

In [6]: !pip install scikit-learn

Requirement already satisfied: scikit-learn in c:\users\naval\anaconda3\lib\site-packages (1.7.2)
 Requirement already satisfied: numpy>=1.22.0 in c:\users\naval\anaconda3\lib\site-packages (from scikit-learn) (2.3.5)
 Requirement already satisfied: scipy>=1.8.0 in c:\users\naval\anaconda3\lib\site-packages (from scikit-learn) (1.16.3)
 Requirement already satisfied: joblib>=1.2.0 in c:\users\naval\anaconda3\lib\site-packages (from scikit-learn) (1.5.2)
 Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\naval\anaconda3\lib\site-packages (from scikit-learn) (3.5.0)

In [7]: !pip install sympy

Requirement already satisfied: sympy in c:\users\naval\anaconda3\lib\site-packages (1.14.0)
 Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\naval\anaconda3\lib\site-packages (from sympy) (1.3.0)

Machine Problem 002: Inverse Laplace Transforms

Finding a function $f(t)$, given the Laplace transform $F(s)$. If $Lf(t) = F(s)$ we write:

$$f(t) = \mathcal{L}^{-1}(F(s))$$

The SymPy library is a symbolic mathematics library written for Python that can be used to perform analytical calculations. This library can be very useful for control system design and analysis.

Table of Common Laplace Transforms

Mathematicians have developed tables of commonly used Laplace transforms. Here is summary table with a few of the entries that will be most common for analysis of linear differential equations.

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at}u(t)$	$\frac{1}{s+a}$
6.	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos(\omega t)u(t)$	$\frac{s}{s^2 + \omega^2}$

```
In [8]: import sympy as sym
from sympy.abc import s,t,u,x,y,z,a,b,c,k
from sympy.integrals import inverse_laplace_transform
```

```
In [9]: # For pretty printing
sym.init_printing()
```

```
In [10]: # Define variables for time 't' and complex variable 's'
# t = sym.symbols('t', real=True)
# s = sym.symbols('s')

# Define a constant:
# a = sym.symbols('a', real=True, positive=True)
```

1. Evaluate:

$$\mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\} = t^2$$

```
In [11]: # Define the frequency domain function:
f1s = 2/s**3
f1s
```

```
Out[11]:  $\frac{2}{s^3}$ 
```

```
In [12]: # Compute the Inverse Laplace Transform:
f1 = inverse_laplace_transform(f1s, s, t)
f1
```

```
Out[12]:  $t^2\theta(t)$ 
```

2. Evaluate:

$$\mathcal{L}^{-1} \frac{s}{s^2 + 1} = \cos(t)$$

```
In [13]: # Define the frequency domain function:
f2s = s/(s**2+1)
f2s
```

```
Out[13]:  $\frac{s}{s^2 + 1}$ 
```

```
In [14]: # Compute the Inverse Laplace Transform:
f2 = inverse_laplace_transform(f2s, s, t)
f2
```

```
Out[14]:  $\cos(t)\theta(t)$ 
```

3. Evaluate:

$$\mathcal{L}^{-1} \frac{1}{s^2 + 1} = \sin(t)$$

```
In [15]: # Define the frequency domain function:
f3s = 1/(s**2+1)
f3s
```

```
Out[15]:  $\frac{1}{s^2 + 1}$ 
```

```
In [16]: # Laplace transform (t->s)
f3 = inverse_laplace_transform(f3s, s, t)
f3
```

```
Out[16]:  $\sin(t)\theta(t)$ 
```

4. Evaluate:

$$\mathcal{L}^{-1} \frac{10}{(s+2)^4} = \frac{5t^3 e^{-2t}}{3}$$

In [17]: *# Define the frequency domain function:*

In [18]: *# Compute the Inverse Laplace Transform:*

5. Evaluate:

$$\mathcal{L}^{-1} \frac{s+1}{(s+1)^2+4} = e^{-t} \cos 2t$$

In [19]: *# Define the frequency domain function:*

In [20]: *# Compute the Inverse Laplace Transform:*

6. Evaluate:

$$\mathcal{L}^{-1} \frac{15}{(s-1)^2-9} = 5e^t \sinh 3t$$

In [21]: *# Define the frequency domain function:*

In [22]: *# Compute the Inverse Laplace Transform:*

7. Evaluate:

$$\mathcal{L}^{-1} \frac{4s-1}{s^2-s} = 1 + 3e^t$$

In [23]: *# Define the frequency domain function:*

In [24]: *# Compute the Inverse Laplace Transform:*